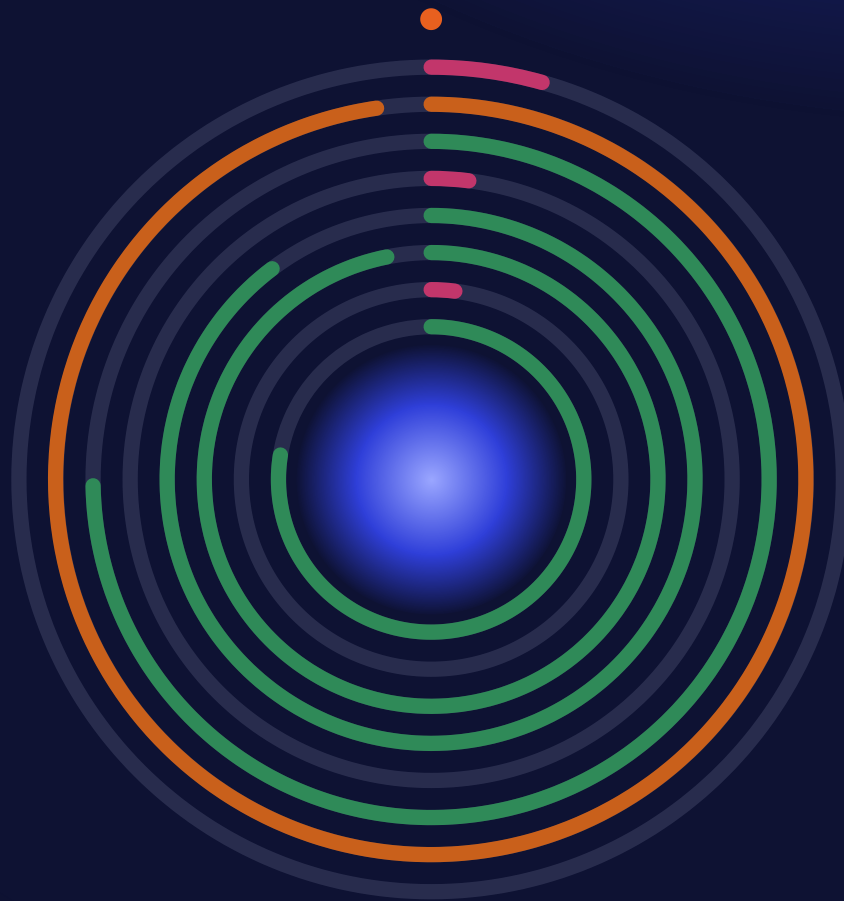


LLM Release Claims Audit

Do vendor claims survive independent measurement?



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25 Sept 2026

Each ring is one claim or stress test; arc length shows how much of it held

What we checked, against what evidence, and how to read the verdicts.

Scope

4 model releases from 24 Nov 2025 to 22 Sept 2026: 9 vendor claims, 2 Assay stress tests and 1 test hypothesis. Vendor claims are paraphrased from each vendor's announcement; stress tests re-check a claim outside the conditions the vendor stated, and are never counted against the vendor.

Evidence base

19 sources: 4 vendor announcements and pricing pages, 9 independent measurements, and 6 secondary reports used where no primary figure was available. Every number carries a numbered citation.

Verdict key

How much of the claimed improvement independent data confirms

Hallmarked verified	Plated partly true	Fool's gold marketing	Unassayed no public data
Delivered at least 80% of the claim.	Delivered 40–79%, or delivered in full but measured quality fell.	Under 40% of the claim, or it got worse.	No independent measurement yet. Run the live pack.

Solid badges are vendor claims. Outlined badges are Assay stress tests (marked Held or Reversed) and hypotheses; they are never counted against the vendor.

Contents

1. Executive summary
2. The evidence at a glance
3. Market context
4. Anthropic: Claude Opus 5.5 against Opus 5
5. OpenAI: GPT-6 Sol and Luna against GPT-5.6
6. Google: Gemini 3.8 Flash against 3.7 Flash
7. Anthropic: Claude Opus 4.5 token efficiency against Sonnet 4.5
8. Methodology and limitations
9. Sources

4 of 6 vendor claims held up on independent data; 1 held only in part and 1 fell short.

Both savings we re-tested beyond the vendors' stated settings reversed: the gains are real, but only where the vendor said they would be.

9

vendor claims examined across 4 releases

67%

of vendor claims with independent data held up in full

2 of 2

stress tests where the saving reversed outside the stated conditions

Exhibit 1

Verdicts on vendor claims



Vendor claims only; stress tests and hypotheses are reported separately. Source: Assay analysis of vendor announcements against independent measurements, cited in each chapter.

Key findings

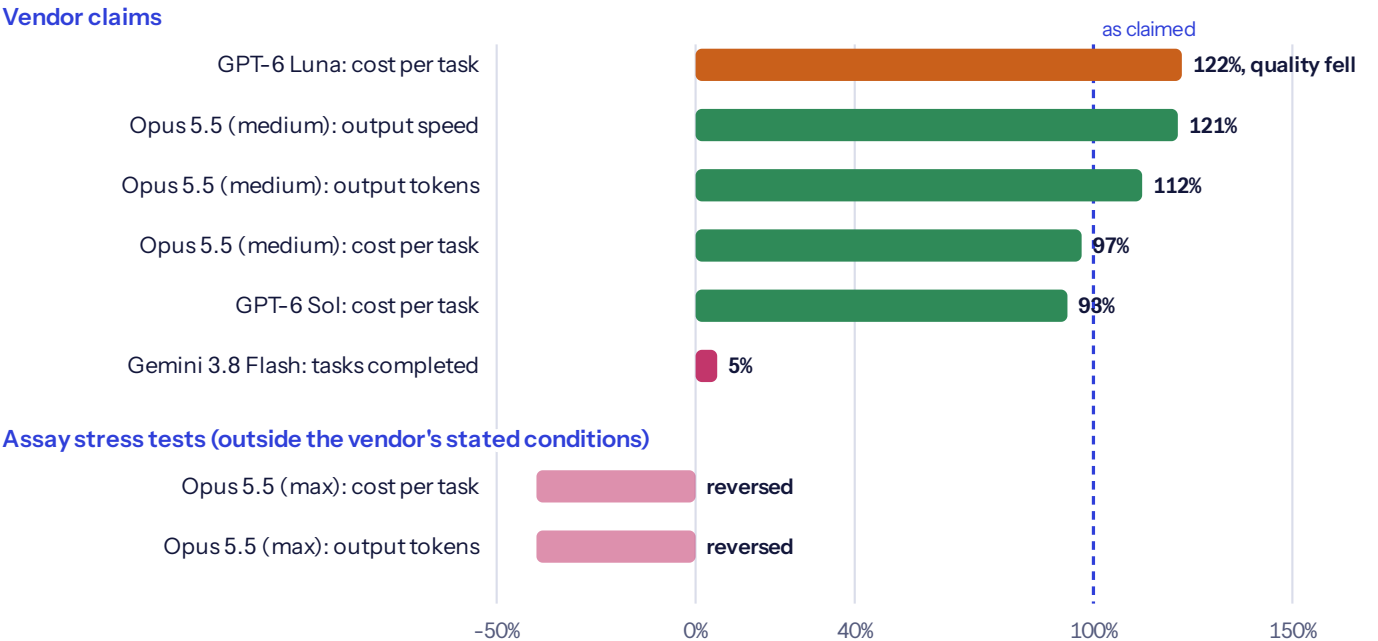
- **Anthropic, Opus 5.5 (medium): claim held.** Costs about 40% less to run than Opus 5 on typical workloads at default effort. Claimed 40% cheaper; independent data shows 39% cheaper.^[2, 3]
- **OpenAI, GPT-6 Luna: held in part.** GPT-6 Luna cuts prices by about half. Claimed 50% cheaper; independent data shows 61% cheaper, with lower measured quality.^[9]
- **Google, Gemini 3.8 Flash: fell short.** Completed more than three times as many tasks as 3.7 Flash. Claimed 200% higher score; independent data shows 11% higher score.^[15]
- **Settings matter.** The same 40% saving, checked at max effort: 2% more expensive. This is outside what the vendor claimed, but it is where an unpinned configuration can land.^[4, 5]
- **3 vendor claims have no independent data yet.** Live test packs are ready so they can be measured on your own workload.

What leaders should do

Test at the effort level you will actually deploy, budget on cost per correct answer rather than price per token, treat any single-benchmark claim as a hypothesis until it survives a noise check, and re-run the audit on every release.

Savings hold at the settings vendors quote; push the settings further and some reverse.

Exhibit 2
Share of each claimed improvement confirmed by independent data



Bars are coloured by verdict; stress tests are shaded lighter. "Quality fell" marks a saving that came with a lower benchmark score. Source: independent measurements cited in each chapter; Assay analysis.

Exhibit 3
Verdict matrix by claim type

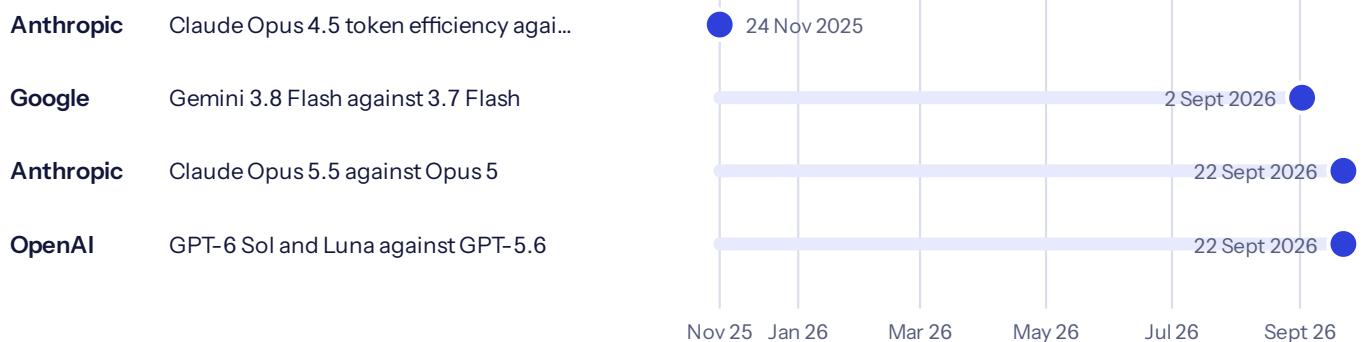
Release	Cost	Speed	Tokens	Quality
Anthropic Claude Opus 5.5 against Opus 5	Hallmarked Reversed	Hallmarked	Hallmarked Reversed	-
OpenAI GPT-6 Sol and Luna against GPT-5.6	Hallmarked Plated	-	-	Unassayed
Google Gemini 3.8 Flash against 3.7 Flash	-	-	Unassayed	Fool's gold
Anthropic Claude Opus 4.5 token efficiency against Sonnet 4.5	Unassayed	-	Unassayed	-

Outlined badges are Assay stress tests or hypotheses, not vendor claims.

3 of 4 upgrades cut list prices, by 20% to 58% on output; 1 held its price.

Exhibit 4

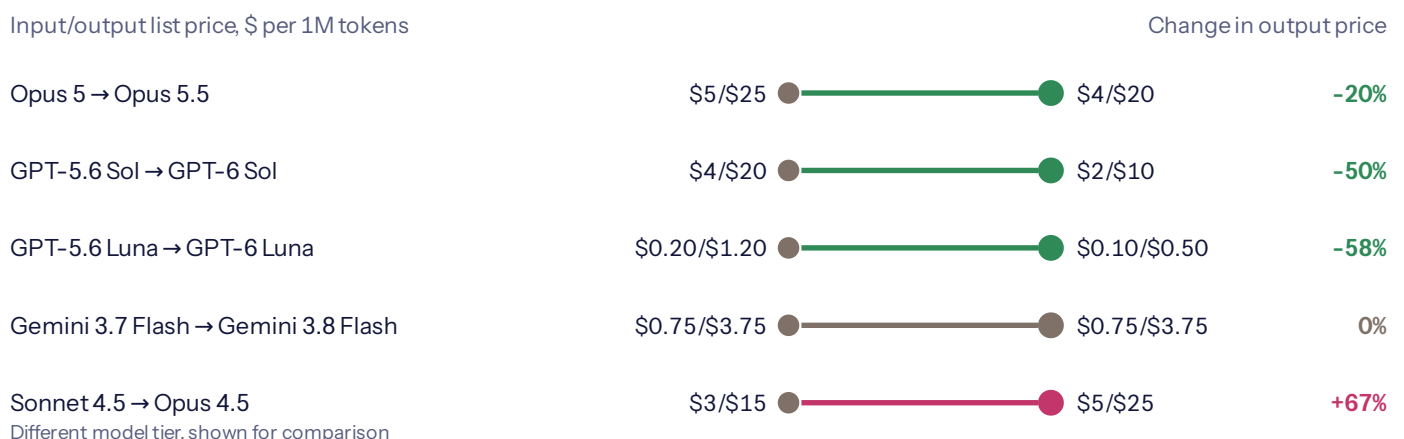
Release timeline



Blue: reviewed scenarios. Orange: drafts awaiting review.

Exhibit 5

List price per release



List prices at release, from the vendor pricing pages and announcements cited in each chapter. A lower price per token lowers cost per task only if token use does not rise.

Claude Opus 5.5 against Opus 5

Released 22 Sept 2026. Sources checked 25 Sept 2026.

In brief

Anthropic cut Opus prices by 20% and says Opus 5.5 costs about 40% less to run than Opus 5 at default effort, while generating output more than 30% faster. Independent measurements support the claim at the default medium effort. At max effort the story reverses: Opus 5.5 writes far more tokens and costs about the same per task.

Claims examined

- Costs about 40% less to run than Opus 5 on typical workloads at default effort^[1]
- Generates output more than 30% faster than Opus 5^[1]
- Uses fewer tokens per task (customer reports range from 20% to 66% fewer)^[1]

Our additional checks

- The same 40% saving, checked at max effort (stress test)
- Fewer tokens per task, checked at max effort (stress test)

Models and list prices

Model	Input	Output
Opus 5 (medium) claude-opus-5	\$5.00	\$25.00
Opus 5.5 (medium) claude-opus-5-5	\$4.00	\$20.00
Opus 5 (max) claude-opus-5	\$5.00	\$25.00
Opus 5.5 (max) claude-opus-5-5	\$4.00	\$20.00

\$ per 1M tokens. Source:^[1]

So what

The claim holds under the vendor's stated conditions but not outside them. Pin the setting you deploy in your contract or configuration, and test at that setting.

Exhibit 6

What independent data shows

Hallmarked

97% of claim

Costs about 40% less to run than Opus 5 on typical workloads at default effort^[1]

Anthropic. Claimed: 40% cheaper. Independent data: 39% cheaper.



Published figures: \$2.19 → \$1.34 per index task. Artificial Analysis Intelligence Index v4.3.2, both at medium effort.^[2, 3]

Reversed

outside stated terms

The same 40% saving, checked at max effort Assay stress test

Anthropic. Claimed: 40% cheaper. Independent data: 2% more expensive.



Published figures: \$5.86 → \$5.98 per index task. Artificial Analysis Intelligence Index v4.3.2, both at max effort. ^[4, 5] Anthropic's claim is for default effort. This row shows what happens if you run both at max.

Hallmarked

121% of claim

Generates output more than 30% faster than Opus 5^[1]

Anthropic. Claimed: 30% faster. Independent data: 36% faster.



Published figures: 55 → 75 output tokens per second. Measured on Anthropic's API at medium effort. ^[2, 3]

Hallmarked

112% of claim

Uses fewer tokens per task (customer reports range from 20% to 66% fewer)^[1]

Anthropic, quoting customers. Claimed: 20% fewer tokens. Independent data: 22% fewer tokens.



Published figures: 49 → 38 million tokens across the index. Total tokens generated on the index at medium effort. ^[2, 3] Tested against the lowest customer figure, 20%.

Reversed

outside stated terms

Fewer tokens per task, checked at max effort Assay stress test

Anthropic, quoting customers. Claimed: 20% fewer tokens. Independent data: 63% more tokens.



Published figures: 73 → 119 thousand output tokens per task. Roughly 1.6 times the output tokens per task. ^[6] The token claim is not specific to max effort. This row shows the setting where it breaks.

What else the data shows

- Per-token prices fell 20% (\$5/\$25 to \$4/\$20) and cache reads fell 60% (\$0.50 to \$0.20), so part of the saving is pricing, not efficiency. ^[1, 6]
- Thinking can no longer be switched off on Opus 5.5, so pipelines that ran Opus 5 without thinking lose that cheaper option. ^[1]

Live test pack not yet run for this release.

GPT-6 Sol and Luna against GPT-5.6

Released 22 Sept 2026. Sources checked 25 Sept 2026.

In brief

OpenAI halved token prices for its mid and low tiers. Independent measurement confirms cost per task roughly halves, driven entirely by price: both models use slightly more output tokens than before, and Luna's benchmark score dropped two points.

Claims examined

- GPT-6 Sol costs half as much as GPT-5.6 Sol^[8]
- GPT-6 Luna cuts prices by about half^[8]
- Sol makes about half as many factual mistakes as GPT-5.6 Sol^[8]

Models and list prices

Model	Input	Output
GPT-5.6 Sol gpt-5.6-sol	\$4.00	\$20.00
GPT-6 Sol gpt-6-sol	\$2.00	\$10.00
GPT-5.6 Luna gpt-5.6-luna	\$0.20	\$1.20
GPT-6 Luna gpt-6-luna	\$0.10	\$0.50

\$ per 1M tokens. Source:^[8]

So what

The saving is real, but part of it came with lower measured quality. Check your own quality bar before switching.

Exhibit 7

What independent data shows

Hallmarked

93% of claim

GPT-6 Sol costs half as much as GPT-5.6 Sol^[8]

OpenAI. Claimed: 50% cheaper. Independent data: 47% cheaper.



Published figures: \$1.99 → \$1.06 per index task. Artificial Analysis Intelligence Index v4.3.2, both at max effort.^[9] OpenAI states the cut per token; this checks it per completed task.

Plated

122% of claim

GPT-6 Luna cuts prices by about half^[8]

OpenAI. Claimed: 50% cheaper. Independent data: 61% cheaper.



Published figures: \$0.18 → \$0.07 per index task. Artificial Analysis Intelligence Index v4.3.2, both at max effort.^[9]

Intelligence Index fell from 43 to 41, so the gain came with lower measured quality.

Unassayed

no public data

Sol makes about half as many factual mistakes as GPT-5.6 Sol^[8]

OpenAI. No figure that can be tested directly.

OpenAI's figure comes from its own set of user-flagged conversations. Artificial Analysis reports its knowledge and hallucination index rising from 22 to 27, which points the same way but measures something different. Test it with a factual task set of your own.

What else the data shows

- The saving comes from price: output tokens per task rose slightly, from about 29k to 31k for Sol and 41k to 51k for Luna.^[10]
- Independent evals show regressions too: Sol dropped about 100 Elo on GDPval-AA v2.1, a benchmark of economically valuable tasks.^[10]
- Prompts above 272K input tokens are billed at higher long-context rates, so very long-context workloads save less.^[11]

Live test pack not yet run for this release.

Gemini 3.8 Flash against 3.7 Flash

Released 2 Sept 2026. Sources checked 25 Sept 2026.

In brief

Same price, same limits, three weeks apart. Google leads with a customer saying 3.8 Flash completed more than three times as many tasks. Independent indexes show a real but much smaller gain in agentic work, flat coding, and a slower first token.

Claims examined

- Completed more than three times as many tasks as 3.7 Flash^[15]
- Agentic video understanding uses up to 88% fewer tokens on long-form content^[14]

Models and list prices

Model	Input	Output
Gemini 3.7 Flash gemini-3.7-flash	\$0.75	\$3.75
Gemini 3.8 Flash gemini-3.8-flash	\$0.75	\$3.75

\$ per 1M tokens. Source:^[14]

So what

The headline figure comes from a single customer's evaluation; independent data shows a much smaller gain. Treat it as an upper bound and measure on your own workload.

Exhibit 8

What independent data shows

Fool's gold
5% of claim

Completed more than three times as many tasks as 3.7 Flash^[15] Proxy measure
Customer quote on Google's Flash page. Claimed: 200% higher score. Independent data: 11% higher score.

no change

Measured 11% higher score

Claimed 200% higher score

Published figures: 45.1 → 50 Agentic Index points. Artificial Analysis Agentic Index.^[15] The quote comes from one company's document-heavy evaluation. The reference is an independent agentic index, so it is a proxy, not a replication.

Unassayed
no public data

Agentic video understanding uses up to 88% fewer tokens on long-form content^[14]
Google. Claimed: 88% fewer tokens.
No independent measurement found. Build a long-video task set and run it live; note the claim says up to 88%, so a typical figure will be lower.

What else the data shows

- Google itself cautions that the model may use more tokens to maximise performance, especially at higher effort levels, so cost per task can rise even at an unchanged price.^[16]
- Time to first token got worse, from 12.0 to 13.2 seconds, even though streaming speed rose from 279 to 299 tokens per second.^[15]
- The independent coding index barely moved (76.1 to 76.3), while the overall Intelligence Index rose from 56 to 59.^[15]

- Both models are on introductory pricing of \$0.75/\$3.75 that rises to \$1.50/\$7.50 on 1 January 2027.^[17]

Live test pack not yet run for this release.

Claude Opus 4.5 token efficiency against Sonnet 4.5

Released 24 Nov 2025. Sources checked 25 Sept 2026.

In brief

A widely repeated efficiency claim from late 2025 that was measured on one benchmark under the vendor's own harness. There is no matching independent number, so this scenario is a live test pack only.

Claims examined

- At medium effort, matches Sonnet 4.5 on SWE-bench Verified with 76% fewer output tokens^[18]

Our additional checks

- Cheaper per task than Sonnet 4.5 despite higher token prices (hypothesis)

Models and list prices

Model	Input	Output
Sonnet 4.5 claude-sonnet-4-5	\$3.00	\$15.00
Opus 4.5 (medium) claude-opus-4-5-20251101	\$5.00	\$25.00

\$ per 1M tokens. Source:^[18]

So what

There is no independent evidence yet. Run the live pack before relying on these numbers.

Exhibit 9

What independent data shows

Unassayed
no public data

At medium effort, matches Sonnet 4.5 on SWE-bench Verified with 76% fewer output tokens^[18]

Anthropic. Claimed: 76% fewer tokens.
The figure is from one coding benchmark on Anthropic's harness. The live pack tests whether the saving carries over to short, general tasks, where it is likely to be smaller.

Unassayed
no public data

Cheaper per task than Sonnet 4.5 despite higher token prices Assay hypothesis

Implied by the token and price claims. Claimed: 20% cheaper.
Opus 4.5 costs \$5/\$25 against Sonnet 4.5's \$3/\$15, so it only wins per task if it really writes far fewer tokens.
The 20% target here is illustrative.

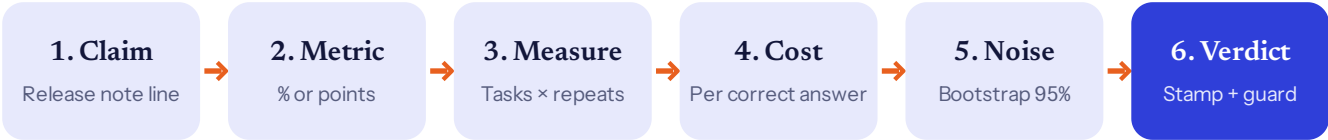
What else the data shows

- At high effort, Anthropic says Opus 4.5 beat Sonnet 4.5 by 4.3 points while using 48% fewer tokens, so the saving shrinks as effort rises.^[18]

Live test pack not yet run for this release.

Every verdict follows the same six steps, and every number traces back to a source.

Exhibit 10
The Assay validation pipeline



Rule, applied in order	Verdict
Live runs only: the 95% bootstrap range crosses zero	Unassayed
The measure moved the wrong way	Fool's gold
80% or more of the claimed improvement delivered	Hallmarked
40% to 79% delivered, or 80%+ delivered while accuracy or the benchmark index fell by 2 points or more	Plated
Under 40% delivered	Fool's gold
No independent measurement exists	Unassayed

Improvements are relative for cost, tokens, latency and speed, and absolute points for accuracy and benchmark indexes. Cost per task counts input tokens and all output tokens, including hidden reasoning, at list price. Vendor claims are tested as stated; stress tests re-run a claim outside the vendor's stated conditions and are reported separately. Live runs grade every answer automatically, repeat each task, and use a paired bootstrap over tasks with 800 resamples. Reference verdicts compare two published figures and carry no noise range.

Limitations

Independent benchmarks use their own task mixes, harnesses and effort settings, which may not match yours. Where no primary figure was published, secondary reporting is cited and labelled as such. Model IDs marked for confirmation, and draft scenarios extracted automatically from release notes, should be reviewed before decisions are made. This report was generated by Assay on 25 Sept 2026.

References

Vendor: the company's own announcement or pricing page. Independent: a third party that ran its own measurements. Secondary: reporting that summarises either.

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